



IoT Project — Task Brief

Submission Template for Participants

Project: AgriSmart — IoT Smart Irrigation for Egyptian Farmers

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Section 1 — Problem Validation

What is the problem?

Small and medium-scale Egyptian farmers make irrigation timing and volume decisions — when to irrigate, and for how long — without real-time, quantified information about the actual condition of the soil at the root zone. Decisions are made instead on fixed schedules, surface-level visual inspection, and personal experience. None of these three inputs reflect subsurface soil moisture, temperature, or salinity at the moment the decision is made. The result is a structural information gap, not a competence gap: the farmer is not missing knowledge of how to irrigate — they are missing the data needed to apply that knowledge precisely.

Who experiences it?

Small and medium-scale farmers who irrigate manually or on a fixed rotation, particularly those relying on surface (flood/furrow) irrigation, which is the dominant method in Egyptian agriculture. Egyptian landholding is heavily fragmented into smallholdings: national agricultural census data shows the average holding size fell from 3.8 feddans (1960) to 2.2 feddans (2010), and the share of holdings under one feddan rose from 26.4% to 48.3% over the same period. The Food and Agriculture Organization defines Egyptian smallholders as those farming under three feddans, and roughly 3.7 million smallholder families farm about 40% of the country's agricultural land on that basis.

Why is it important?

Agriculture consumes more than 85% of Egypt's Nile water allocation, while national per-capita renewable water availability has fallen to approximately 628 m³/year — below the internationally recognized water-scarcity threshold. Field-level irrigation application efficiency remains around only 66%, meaning roughly a third of applied irrigation water is not effectively used by the crop, at precisely the point in the system (the field itself) that AgriSmart instruments. This is not a marginal inefficiency: the Egyptian government has committed an estimated EGP 900 billion (2017–2037) to a national water strategy that names irrigation modernization as one of four core pillars, specifically because on-farm water use is treated as a lever the country cannot afford to leave unaddressed.

Evidence (Interview, Survey, Observation, Research)

The evidence above is drawn from Egyptian agricultural census data (Ministry of Agriculture and Land Reclamation, 1960/1990/2010), FAO's Family Farming Knowledge Platform, a peer-reviewed Nature Communications study on Egypt's water balance, Fanack Water's water-use reporting, and Egyptian

government statements on the National Water Resources Plan. Full citations are in the accompanying evidence document.

Data gap — flagged, not fabricated: No farmer interviews, surveys, or field observations have been conducted for this submission. Every claim above is desk research against secondary/institutional sources, not primary customer evidence. A structured interview instrument is designed and ready to run (see Section 2), but no results exist yet and none are presented here as if they did.

Section 2 — Target Customer & Named Client

Customer Segment

Small and medium-scale farmers in the Egyptian countryside (Nile Valley/Delta and reclaimed New Land) who personally make day-to-day irrigation decisions on their own land, using surface or basic sprinkler irrigation, without existing precision-agriculture instrumentation. Based on national census data, this segment is best characterized as farmers operating holdings in the low single-digit feddan range — the modal Egyptian holding is well under 3 feddans, and nearly half of all holdings nationally are under one feddan.

Named Client

Data gap — flagged, not fabricated: No named client, pilot farm, or design partner has been secured. Presenting a fabricated individual or farm here would violate the instruction not to invent customers, so this field intentionally states the target profile only, not an identified person or business. Before submission, you should either (a) name a real farmer, farm, or cooperative contact you have already spoken with, or (b) leave this as an explicit customer-discovery target and say so plainly to the judges — both are defensible; inventing a name is not.

Target profile for the discovery process described below: an owner-operator farmer cultivating a small-to-medium field-crop or horticulture holding, who personally decides irrigation timing, and who has access to a smartphone.

Key Pain Points

- Irrigation timing uncertainty without root-zone data
- Risk of over-irrigation from a "better safe than sorry" habit, on a national resource already in deficit
- Invisible, gradually accumulating soil salinity risk
- No historical, season-over-season irrigation record
- Recurring time cost of manually walking fields to inspect soil condition
- Exclusion by existing precision-agriculture tools that are not Arabic-first or priced for a smallholder

Data gap — flagged, not fabricated: These pain points are hypotheses derived from the problem-validation evidence in Section 1, not confirmed farmer statements. None has yet been ranked or confirmed by an actual farmer.

Validation Summary

Problem-level validation is evidence-based, drawn from Egyptian census data, FAO, Nature Communications, Fanack Water, and Egyptian government reporting (Section 1). Customer-level validation has not yet been collected. A 12-question interview instrument has been designed to test irrigation decision-making, soil-moisture and salinity awareness, water waste, current irrigation methods,



willingness to use sensors and a mobile app, willingness to automate irrigation, price sensitivity, and trust/usability — intended to run with 10–15 farmers across at least two governorates with different irrigation contexts before pricing or a named client is finalized.

Section 3 — IoT Solution Overview

Solution Description

AgriSmart is an ESP32-based field sensing and irrigation-control unit that measures soil moisture, temperature, and salinity/electrical conductivity at the root zone, and reports this telemetry to a cloud backend surfaced through an Arabic-first mobile application with Egyptian colloquial Arabic voice interaction. Where a farm has an installed solenoid valve, the same application can issue a remote open/close command back to the field unit.

How IoT Is Used

IoT is structurally necessary, not decorative: the missing information in this problem — actual root-zone soil condition — does not exist anywhere digitally until a physical sensor measures it in the ground. A mobile app alone cannot generate that data. Equally, closing the loop by safely executing a farmer-approved irrigation command on physical valve hardware requires a connected device in the field; it cannot be done by software alone. AgriSmart's IoT layer is what makes the sense-decide-act loop possible at all, not an add-on to a software product that could otherwise work without it.

Value Proposition

For small and medium-scale Egyptian farmers who irrigate without reliable, real-time knowledge of their soil's actual condition, AgriSmart is an Arabic-first IoT soil-monitoring and irrigation-control system that reports real-time root-zone moisture, temperature, and salinity to the farmer's phone — including through spoken Egyptian colloquial Arabic — so that irrigation decisions are made from measured ground truth instead of guesswork, on hardware and pricing scoped to a single small farm rather than an enterprise operation.

Main Features

- Real-time soil moisture, temperature, and salinity/EC monitoring at the root zone
- Arabic-first mobile app with Egyptian colloquial voice interaction
- Remote irrigation valve control with tracked command status (sent / acknowledged / executed)
- Historical, per-field soil and irrigation data
- Secure, independently authenticated device identity with farm-scoped access control

Section 4 — System Design

System Architecture Diagram

Telemetry path: Soil → Sensors (moisture / temperature / EC) → ESP32 Edge Device → Connectivity (cellular, with Wi-Fi as an opportunistic fallback where a farm router is in range) → AgriSmart Backend (device authentication, input validation, irrigation safety rules) → Database → Mobile Application → Farmer.

Control path (reverse direction): Farmer → Mobile Application → Backend (command state machine: queued → sent → acknowledged → executing → completed/failed) → Connectivity → ESP32 → Relay → Irrigation Valve → Field. The device's own acknowledgment and execution-result report are the only accepted source of truth for what physically happened — the backend never treats "command sent" as equivalent to "valve moved."

Hardware Components

Hardware	Purpose
ESP32 microcontroller module	Field edge compute: reads sensors, drives the valve relay, manages connectivity and low-power (deep-sleep) duty cycling
Capacitive soil moisture probe	Measures root-zone moisture; avoids the electrode corrosion resistive probes suffer in saline Egyptian soils
Waterproof digital soil temperature probe	Measures root-zone temperature; supports correct interpretation of moisture and EC readings
Soil EC / salinity probe	Measures bulk soil electrical conductivity as a salinity-risk indicator, read jointly with moisture
Relay module + solenoid irrigation valve	Physically opens/closes the water line to the field zone on a validated backend command
Solar panel + rechargeable battery + charge controller	Powers the field unit independently of mains electricity, sized to a low-duty-cycle sampling schedule
IP65/IP67-rated enclosure with sealed cable glands	Protects electronics from dust, humidity, and irrigation-water exposure in the field

Data gap — flagged, not fabricated: Specific component models, unit costs, and total bill-of-materials pricing have not been sourced/quoted and are not stated here. If judges will ask for a per-unit cost, this needs supplier quotes before the pitch, not an estimated figure invented for this document.

Software Components

Software	Purpose
Mobile application (Arabic-first, voice-enabled)	Farmer-facing display of soil data/history; issues irrigation commands; hosts Egyptian colloquial Arabic voice interaction
Backend API (routes / controllers / services / repositories)	Authenticates users and devices, validates all inbound data, exposes farm/device/telemetry/command endpoints

Software	Purpose
Device authentication service	Verifies ESP32 device identity independently of user accounts before accepting telemetry or command acknowledgments
Telemetry ingestion & validation service	Accepts, schema-validates, and timestamps incoming soil readings
Irrigation command service (state machine)	Tracks each command through queued/sent/acknowledged/executing/completed so an issued command is never assumed executed
Authorization layer (farm-scoped, anti-IDOR)	Ensures a farmer can only see or control their own farm's devices and valves
Database	Persists telemetry history, farm/device/valve relationships, and full command audit history

Section 5 — Impact

SDG Alignment

- SDG 2 — Zero Hunger (more resilient crop outcomes through better-informed irrigation)
- SDG 6 — Clean Water and Sanitation (directly targets water-use efficiency in agriculture, the largest consumer of Egypt's water)
- SDG 12 — Responsible Consumption and Production (reduces resource waste at the point of use)
- SDG 13 — Climate Action (supports adaptation to increasing water stress)
- SDG 15 — Life on Land (mitigates soil salinization risk from uninformed over-irrigation)

Expected Impact

At the farmer level: replaces guesswork with a measured basis for irrigation decisions and gives the farmer, for the first time, a personal historical record of how their own field has actually been irrigated. At the farm level: creates the measurement precondition for reducing both over- and under-irrigation on the instrumented zone. At the environmental level: targets the specific mechanism — uninformed over-application on surface-irrigated, salinity-sensitive soils — that current practice cannot avoid without root-zone data. At the national level: directly supports Egypt's own EGP 900 billion National Water Resources Plan (2017–2037), which names irrigation modernization as a core strategic pillar.

Impact will be measured, not assumed, against defined KPIs once piloted: water applied per irrigation cycle and per feddan, irrigation events per crop cycle, soil-moisture stability, farmer time spent on manual monitoring, irrigation decision accuracy, system uptime, and sensor reliability.

Data gap — flagged, not fabricated: No pilot has been run and no water-savings percentage, yield improvement, or other outcome figure is claimed anywhere in this submission. Any specific number (e.g. "X% water saved") must come from an actual pilot before it is used in a pitch — stating one now would be fabricated evidence.

Why Is This Solution Feasible?

The backend's device-authentication, farm-scoped authorization, telemetry-validation, and irrigation command-safety architecture is already implemented and tested — this is not a paper design. The ESP32 platform, the sensor categories specified, and cellular connectivity are standard, mature, commercially available components, which keeps the bill of materials appropriate for a single small-farm MVP rather than requiring unproven technology. The MVP scope is deliberately narrow — single-zone sensing per field, farmer-initiated (not autonomous) control, and no AI/predictive claims — which keeps near-term technical risk low while preserving an architecturally supported path to the predictive features named explicitly as future work, not as already built.

Full evidence log, the customer-discovery interview instrument, hardware/software rationale, connectivity trade-off analysis, and the complete list of flagged data gaps are provided in the accompanying document: [AgriSmart_Task2_Full_Strategy.md/.docx](#).

